

Computer assignment 1

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Exercise 1:1

```
## Warning: package 'epitools' was built under R version 4.5.2
```

The raw data with their sums are as below:

```
##      Opinion
## Gender Favor Against Sum
##  Women   309    191  500
##   Men    319    281  600
##   Sum    628    472 1100
```

1:1.1 Calculate percentages

Calculate the percentage in favor and against legal abortion for men and women separately:

$$P(\text{women-in favor}) = \frac{309}{500} \times 100 = 61.8\%$$

$$P(\text{women-against}) = \frac{191}{500} \times 100 = 38.2\%$$

$$P(\text{men-in favor}) = \frac{319}{600} \times 100 = 53.17\%$$

$$P(\text{men-against}) = \frac{281}{600} \times 100 = 46.83\%$$

1:1.2 Chi-square test and likelihood ratio test

Hypothesis testing

H0: Gender and opinion are independent.

H1: Gender and opinion are not independent.

Expected values:

$$E_{ij} = \frac{\text{Row total} \times \text{Column total}}{\text{Grand total}}$$

$$E_{11} = \frac{500 \times 628}{1100} = 285.45, \quad E_{12} = \frac{500 \times 472}{1100} = 214.55$$

$$E_{21} = \frac{600 \times 628}{1100} = 342.55, \quad E_{22} = \frac{600 \times 472}{1100} = 257.45$$

Chi-square statistic:

$$\begin{aligned} \chi^2 &= \sum \frac{(O - E)^2}{E} \\ &= \frac{(309 - 285.45)^2}{285.45} + \frac{(191 - 214.55)^2}{214.55} + \frac{(319 - 342.55)^2}{342.55} \\ &\quad + \frac{(281 - 257.45)^2}{257.45} = 8.2979 \end{aligned}$$

$$\text{Degrees of freedom (df)} = (2 - 1)(2 - 1) = 1$$

Likelihood ratio statistic:

$$G^2 = 2 \sum O \ln \left(\frac{O}{E} \right)$$

$$G^2 = 8.3223$$

From results above, we can see that Pearson's χ^2 statistic is 8.2979 and likelihood ratio statistics is 8.3223. Both of them are bigger than $\chi^2_1(0.05) = 3.841459$, so we can reject the null hypothesis, which means from this survey, gender and opinion are not independent.

1:1.3: Odds ratio with 95% confidence interval

$$\widehat{OR} = \frac{n_{11}n_{22}}{n_{12}n_{21}}$$

$$SE = \sqrt{\frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}}$$

```
# Calculate odds
odds_women_favor <- women_favor / women_against
odds_men_favor <- men_favor / men_against

# Odds ratio (women vs men)
odds_ratio <- odds_women_favor / odds_men_favor
cat("Odds ratio (women/men):", round(odds_ratio, 3), "\n")

## Odds ratio (women/men): 1.425

# Standard error of Log odds ratio
se_log_or <- sqrt(1/women_favor + 1/women_against + 1/men_favor + 1/men_against)

# 95% confidence interval for odds ratio
```

```
log_or <- log(odds_ratio)
lower_ci_log <- log_or - 1.96 * se_log_or
upper_ci_log <- log_or + 1.96 * se_log_or

lower_ci_or <- exp(lower_ci_log)
upper_ci_or <- exp(upper_ci_log)
cat("95% CI for odds ratio: (", round(lower_ci_or, 3), ",", round(upper_ci_or, 3), ")\n")

## 95% CI for odds ratio: ( 1.119 , 1.814 )
```

Interpretation: Women have 1.43 times the odds of being in favor of abortion compared to men, with 95% confidence interval (1.12, 1.81) indicating that we are 95% confident that in the general population, women's odds of supporting abortion are 12% to 81% higher than men's. Since the confidence interval does not contain $\theta = 1$, the association between gender and opinion is significant.

1:1.4: Risk ratio with 95% confidence interval

$$\hat{r} = \frac{n_{11}/n_1}{n_{21}/n_2}$$

$$SE = \sqrt{\widehat{Var}[\log(\hat{r})]} = \sqrt{\frac{1 - n_{11}/n_1}{n_{11}} + \frac{1 - n_{21}/n_2}{n_{21}}}$$

```
# Calculate risks
risk_women <- women_favor / total_women
risk_men <- men_favor / total_men

# Risk ratio (women vs men)
risk_ratio <- risk_women / risk_men
cat("Risk ratio (women/men):", round(risk_ratio, 3), "\n")

## Risk ratio (women/men): 1.162

# Standard error of log risk ratio
se_log_rr <- sqrt((1/women_favor - 1/total_women) + (1/men_favor - 1/total_men))

# 95% confidence interval for risk ratio
log_rr <- log(risk_ratio)
lower_ci_log_rr <- log_rr - 1.96 * se_log_rr
upper_ci_log_rr <- log_rr + 1.96 * se_log_rr

lower_ci_rr <- exp(lower_ci_log_rr)
upper_ci_rr <- exp(upper_ci_log_rr)
cat("95% CI for risk ratio: (", round(lower_ci_rr, 3), ",", round(upper_ci_rr, 3), ")\n")
```

```
## 95% CI for risk ratio: ( 1.05 , 1.287 )
```

Interpretation: Women are 1.16 times as likely to be in favor of abortion compared to men, with 95% confidence interval (1.05, 1.29) which means that we are 95% confident that women are between 1.05 and 1.29 times as likely as men to favor abortion, and this difference is statistically significant because the interval does not include 1.

1:1.5: Using built-in R functions

#Generate a frequency table and calculate row percentage

```
tab1<- as.table(rbind(c(309, 191), c(319, 281)))
dimnames(tab1) <- list(gender = c("women", "men"),opinion = c("favor", "against"))
addmargins(tab1)
```

```
##          opinion
## gender  favor against  Sum
##  women    309    191  500
##   men    319    281  600
##   Sum    628    472 1100
```

```
addmargins(prop.table(tab1,1),2)
```

```
##          opinion
## gender      favor    against      Sum
##  women 0.6180000 0.3820000 1.0000000
##   men   0.5316667 0.4683333 1.0000000
```

#Calculate X2, G2 and p-values

```
chisq.test(tab1,correct=FALSE)
```

```
##
## Pearson's Chi-squared test
##
## data:  tab1
## X-squared = 8.2979, df = 1, p-value = 0.003969
```

```
loglm(~gender+opinion,tab1)
```

```
## Call:
## loglm(formula = ~gender + opinion, data = tab1)
##
## Statistics:
##              X^2 df      P(> X^2)
## Likelihood Ratio 8.322320  1 0.003916088
## Pearson          8.297921  1 0.003969048
```

#Calculate risk/odds ratio and confidence interval

```
oddsratio(tab1, method = "wald", rev="both")
```

```
## $data
##      opinion
## gender  against favor Total
##  men      281    319    600
##  women    191    309    500
##  Total    472    628   1100
##
## $measure
##      odds ratio with 95% C.I.
## gender  estimate    lower    upper
##  men     1.000000      NA      NA
##  women   1.425085  1.119482  1.814113
##
## $p.value
##      two-sided
## gender  midp.exact fisher.exact  chi.square
##  men           NA           NA           NA
##  women 0.003990219  0.004071121 0.003969048
##
## $correction
## [1] FALSE
##
## attr(,"method")
## [1] "Unconditional MLE & normal approximation (Wald) CI"
```

```
riskratio(tab1,rev="both")
```

```
## $data
##      opinion
## gender  against favor Total
##  men      281    319    600
##  women    191    309    500
##  Total    472    628   1100
##
## $measure
##      risk ratio with 95% C.I.
## gender  estimate    lower    upper
##  men     1.000000      NA      NA
##  women   1.162382  1.049744  1.287107
##
## $p.value
##      two-sided
## gender  midp.exact fisher.exact  chi.square
##  men           NA           NA           NA
##  women 0.003990219  0.004071121 0.003969048
##
## $correction
## [1] FALSE
##
```

```
## attr(,"method")
## [1] "Unconditional MLE & normal approximation (Wald) CI"
```

Compared with “by hand” calculation, all the result are same as the result generated by r package.

Exercise 1:2

1:2.1 Calculate X^2 , G^2 , Odds ratio and Risk ratio with 95% CI

The given data with some sums are as below:

```
##      result
## gender  admitted not_admitted  Sum
## men      1198      1493 2691
## women     557      1278 1835
## Sum      1755      2771 4526
```

Hypothesis testing H_0 : Gender and admission result are independent. H_1 : Gender and admission result are not independent.

By using R functions, we can obtain Pearsons X^2 , likelihood ratio G^2 , odds ratio and risk ratio with confidence interval as shown in the table below:

Results of raw data

Statistic	Value
Counts(W_adm, W_rej; M_adm, M_rej)	557, 1278; 1198, 1493
Total N	4546
Women admit rate	30.4%
Men admit rate	44.5%
Pearson's $X^2(df=1)$; p	92.205; 0
Likelihood-ratio $G^2(df=1)$; p	93.449; 0
OR(M vs W); 95%CI	1.841; [1.624, 2.087]
RR(M vs W); 95%CI	1.466; [1.352,1.591]

Interpretation of the result:

From results above, we can see that Odds Ratio=1.841 and the 95% CI for OR being [1.624, 2.086] doesn't contain $\theta = 1$, so the association between gender and admission result is significant, which means that male applicants were 1.84 times as likely to be admitted as female applicants.

Moreover, Risk Ratio = 1.466 indicates that the probability of getting admitted for male applicants was about 1.47 times that of female applicants, with 95% confidence intervals [1.352,1.591].

Both measures suggest that male applicants had a higher chance of being admitted than female applicants in 1975 and all the p-values are less than 0.05, showing that this difference is statistically significant.

1:2.2 replace all numbers to one tenth and one hundredth

The second task asks us to replace all numbers in the table to one-tenth and one-hundredth. After replacing all the numbers, again we use R-functions to get the statistics listed. The outcome is summarized in the following table.

Results of divided data

Statistic	Value_tenth	Value_hundredth
Counts(W_adm, W_rej; M_adm, M_rej)	56, 128; 120, 149	6, 13; 12, 15
Total N	453	46
Women admit rate	30.4%	31.6%
Men admit rate	44.6%	44.4%
Pearson's $\chi^2(df=1)$; p	9.241; 0.00237	0.774; 0.379
Likelihood-ratio $G^2(df=1)$; p	9.364; 0.00221	0.783; 0.376
OR(M vs W); 95%CI	1.841; [1.240, 2.734]	1.733; [0.507, 5.928]
RR(M vs W); 95%CI	1.466; [1.135, 1.893]	1.407; [0.642, 3.085]

Observation1: After replacing all numbers to one tenth, the value of χ^2 and G^2 also decrease to approximately one-tenth of the original values, and the corresponding p-values get larger. We can also see that the OR and RR values stay almost the same with the values obtained in Question 1. And for the 95% confidence intervals of OR and RR, the range become larger.

Observation2: When the table values are reduced to one-hundredth of the original counts, the χ^2 and G^2 value also decrease by two orders of magnitude, but the slight numerical differences arise from rounding errors introduced when dividing the cell counts by 100; In contrast, the OR and RR values maintain the same order of magnitude as the results from question 1, showing only minor numerical variations. And for the 95% confidence intervals of OR and RR, they show a much wider interval.

Comment on the observation:

The test statistics χ^2 and G^2 are directly proportional to the sample size, because they measure the absolute value of discrepancy between observed and expected values, and with df remaining the same and total n smaller, p-value gets bigger naturally;

OR and RR values are independent of sample sizes, because they value the relative difference based on proportions, as long as the proportion remain the same within

tables, they stay invariant. Moreover, the confidence intervals of OR and RR become wider when sample sizes decrease due to an increase in standard error, reflecting a lower precision of the estimation.

These patterns remind us that when evaluating statistical significance, we should pay attention to both inner proportion and size of a sample. We should always consider the magnitude of the effect along with its precision of estimate, rather than only focusing on p-values.

1:2.3

We assume that $H_0: \theta = 1$. Our aim is to make the population odds ratio (θ) differ from 1, thus we need to reject H_0 , which means a small p-value and a big χ^2 is what we want. From previous result, we know that the larger sample size is, the larger χ^2 we can obtain. Also we need the sample odds ratio to be around 1, thus the new two-way table can be made like below:

```
##      result
## gender  admitted not_admitted    Sum
##   men     500000      500000 1000000
##   women  498550      502900 1001450
##   Sum    998550      1002900 2001450
```

Here we capture some key statistics about this self-made data in the following table:

Results of self-made data

Statistic	Value_hundredth
population odds ratio(under H_0)	1
population odds ratio(under H_1)	$\neq 1$
Sample odds ratio	1.008725, within the interval(0.99,1.01)
p-value	0.00212218

Comment about the relevance to declare ‘statistical significance’:

In this case, we created a two-way table with a sample odds ratio of approximately 1.01, but the hypothesis test showed a statistically significant result ($p < 0.05$), indicating that the population odds ratio is significantly different from 1. This highlights a critical aspect of statistical significance: it is highly influenced by sample size.

With a large sample size (e.g., Total = 2001450 in our table), even trivial effect sizes—like an odds ratio of 1.01, which represents a negligible association in practice—can become statistically significant. This occurs because statistical tests (e.g., Pearson’s chi-square) are designed to detect even minute deviations from the null hypothesis when sample sizes are large. However, statistical significance alone does not imply practical or clinical importance.

In such situations, declaring “statistical significance” can be misleading if interpreted as evidence of a meaningful effect. It emphasizes the need to complement p-values with measures of effect size (e.g., odds ratio, risk ratio) and their confidence intervals. For instance, in this case, the odds ratio is very close to 1, and its confidence interval would likely be narrow and include values near 1, indicating that the effect is negligible in real-world terms.

Therefore, when reporting results, we would prefer to:

- Report the effect size (e.g., odds ratio) and its confidence interval to provide context on the magnitude and precision of the effect.
- Avoid overemphasizing statistical significance alone, especially in large samples, and instead discuss whether the effect size is meaningful in the given domain.
- Encourage readers to focus on practical significance rather than just p-values.

This exercise underscores the importance of not equating statistical significance with practical relevance, a common pitfall in data analysis.